

SMART INDIA HACKATHON · PROBLEM STATEMENT 26154

Pramaan AI

प्रमाण · Content you can prove.

Project document: problem, solution, users, use cases, features, technology and installation

Problem statement: Gen AI Platform for Automated Content Transformation

Organisation: National Technical Research Organisation (NTRO)

Theme: Blockchain & Cybersecurity · **Category:** Software

Team: [TEAM NAME] · **Version:** 1.0, 30 September 2026

This document explains everything about the project in simple words, so every team member, mentor and judge can understand it without a technical background.

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1. The whole project on one page

The problem

Security agencies like NTRO receive long reports every day, for example about a cyber attack. From each report they must write many different things: an advisory, a short summary for senior officers, slides, social media posts, an infographic and a video script. Today this is done by hand. It is **slow**, it needs **expert writers**, the **facts often do not match** between versions, and tools like ChatGPT **cannot be used** because secret data would go to foreign servers.

Our solution

Pramaan AI is an app installed on the organisation's own computer. It works **fully offline** using **Indian AI models**. The user uploads a report once, ticks the formats they need, and the app creates all of them in minutes, in **22 Indian languages plus English**. A reviewer approves the outputs and signs them with an official **digital signature**. Every document gets a **QR code**, so anyone can check whether it is genuine or fake.

What makes it different

- **Fully offline:** secret reports never leave the computer.
- **Indian AI:** Sarvam (Bengaluru), BharatGen (IIT Bombay) and AI4Bharat (IIT Madras).
- **Truthful:** every sentence can be traced back to the original report, and unsupported sentences are flagged.
- **Safe to publish:** sensitive details such as internal IP addresses are hidden automatically in public posts.
- **Impossible to fake:** signed documents carry a QR code, and a public "Is this real?" page catches fake alerts.

Who uses it

Operators create content, **Reviewers** approve and sign it, **Admins** manage the system, and the **Public** can verify documents without logging in.

How it is installed

One command in the terminal installs the app and downloads the Indian AI models once. For secret offices with no internet, everything comes on a USB drive. After that, the app opens in a normal web browser but runs entirely on the user's own computer.

One-line pitch: One report in → every format out: offline, accurate, in every Indian language, and impossible to fake.

2. The problem, clearly explained

2.1 What organisations deal with every day

Government and security organisations constantly receive information in many forms: threat reports, news articles, research papers, policy documents, advisories, incident reports and announcements. This information is useful only when it reaches the right people, in the right form, quickly.

The same piece of information usually has to be communicated to many different audiences. Each audience needs a different format, tone and level of detail:

Audience	What they need	Example format
IT and security teams	Exact technical details and clear steps	Structured advisory
Senior officers	The key points in two minutes, and the decision needed	Executive summary
Meetings and briefings	A visual story with talking points	Presentation with speaker notes
The general public	Simple warnings and what to do	LinkedIn post, X thread, infographic
Awareness campaigns	Something people will watch	Video script with storyboard
Citizens across India	Information in their own language	Any of the above in Hindi, Tamil, Bengali...

2.2 A real-life example

Imagine a ransomware attack is locking computers in hospitals across several states. A 30-page technical report arrives in the morning. Before the end of the day, the team must produce:

1. a formal advisory for hospital IT teams,
2. a one-page summary for senior officers,
3. a presentation for an emergency meeting,
4. social media posts to warn the public,
5. an infographic for quick sharing,
6. a video script for an awareness campaign,
7. and versions of all of these in several Indian languages.

Done by hand, this can take a team **one to two days**. During a cyber attack, every hour of delay means more damage.

2.3 Why the current way fails

Problem	What happens	Why it matters
Slow	Each format is written separately by hand	Warnings reach people late
Needs experts	Only skilled writers with domain	Small teams get overloaded

Problem	What happens	Why it matters
	knowledge can do it	
Facts do not match	One version says 42 hospitals, another says 24	People lose trust in official information
Language gap	Most material is made only in English	Many citizens are left out
Fake advisories	Criminals send fake “government alerts” with scam links	People are tricked into sharing OTPs or clicking malware

2.4 Why ChatGPT-style tools are not the answer

- **Data leaves the country:** online AI tools send the document to servers run by foreign companies. For secret or sensitive reports this is not allowed.
- **AI can make things up:** these tools sometimes write confident but wrong facts (called “hallucinations”). In official communication this is dangerous.
- **No proof:** a document written by AI cannot prove where it came from or whether someone changed it later.
- **Weaker in Indian languages:** many global tools are much better in English than in Indian languages.

2.5 What NTRO has asked for

NTRO wants an **AI-powered content transformation platform**. In simple words:

- A **dashboard** where an operator gives source content: text, documents, articles, reports, prompts, images, videos or other information.
- The operator **selects one or more output types:** video package, LinkedIn post, Twitter/X post, advisory, infographic, executive summary, presentation.
- The operator can **control settings** such as target audience, tone, language, level of detail, objective and style.
- The platform **understands the content** and **creates every selected output** from the same source.

Deliverables for evaluation: source code link, README with setup instructions, architecture document (maximum 2 pages), demo video (maximum 2 minutes) and technical presentation (maximum 5 slides).

3. Our solution: Pramaan AI

3.1 In one sentence

Pramaan AI is an offline app, powered by Indian AI, that turns one report into every format an organisation needs, in 22 Indian languages, while hiding secrets, proving every fact and making every document impossible to fake.

3.2 Why the name

Pramaan (प्रमाण) means “proof” or “evidence”. It describes exactly what makes the app special: every sentence is proved against its source, and every document can be proved genuine with its QR code. The tagline is “Content you can prove.”

3.3 How it works, in 8 steps



Figure 1: The journey of one report through Pramaan AI

- Upload.** The operator gives a report, document, scanned page, image, audio or video, or simply types or speaks a request.
- Safety check.** Before anything else, the app looks for secret markings (like “RESTRICTED”), personal data (like Aadhaar numbers) and hidden tricks meant to fool the AI. It suggests a sharing level (TLP).
- Fact sheet.** The AI reads everything once and writes down the key facts, numbers, dates, affected systems and recommended actions.
- Create.** Every selected output is written from that one fact sheet, so all versions say the same thing. Short outputs appear first.
- Check.** The app traces every sentence back to the source, checks that facts match across all outputs, and checks public posts for panic-causing or unverified wording.
- Approve and sign.** A reviewer checks the work and signs it with a Class 3 digital signature (DSC) token.
- Share.** The signed files are saved as a Campaign Kit, each with its own QR code.
- Verify.** Anyone who receives a document can scan the QR code and see “Genuine” or “Not genuine”.

3.4 The six ideas that make Pramaan AI special

Idea	What it means in simple words	Why judges will care
1. Fully offline	The whole app, including the AI, runs on the organisation’s own computer. No internet is	Secret data never leaves the building. Perfect for a security

Idea	What it means in simple words	Why judges will care
	needed.	agency.
2. Indian AI	Uses models built in India: Sarvam 30B, BharatGen Param2 and AI4Bharat's language tools.	Supports the IndiaAI Mission and Atmanirbhar Bharat.
3. One fact sheet, many outputs	The AI reads the report once and writes every output from the same facts.	No more "42 hospitals" in one version and "24" in another.
4. Every line traced	Click any sentence to see where it came from. Unsupported sentences turn yellow.	Solves the biggest fear about AI: making things up.
5. Safe for the public	Sensitive details are hidden automatically in public posts, and TLP sharing labels are applied.	Prevents accidental leaks.
6. Signed and verifiable	Approved documents are digitally signed, recorded in a tamper-proof record book and carry a QR code.	Fake government advisories become easy to catch.

3.5 How we meet every NTRO requirement

NTRO asked for	How Pramaan AI does it
Dashboard for the operator	Operator dashboard and a 3-step "New transformation" flow
Input as text, documents, articles, reports, prompts	Upload PDF, Word, text, saved web pages; type or speak a prompt
Input as images and videos	Scanned pages read with offline OCR; audio and video turned into text offline
Select one or more output types	7 output tiles; select any combination
Video: script, storyboard, scenes, narration, subtitles, visuals	Full video package plus a narrated preview video with Indian voices and subtitle files
LinkedIn post, Twitter/X post or thread	Both, with character limits checked and sensitive details hidden
Structured advisory	CERT-In style advisory with severity, indicators and actions
Infographic content and layout	Key messages, three layouts and a finished PNG image
Executive summary	One-page briefing with "decision needed", plus an audio version
Presentation slides and speaker notes	Complete PowerPoint deck with notes for every slide
Control audience, tone, language, detail, objective, style	All six controls, plus sharing level and letterhead

NTRO asked for	How Pramaan AI does it
Multiple outputs from the same source	All outputs come from one fact sheet, in parallel

4. Who uses Pramaan AI

Pramaan AI has three kinds of logged-in users and one kind of public visitor. Each logged-in role has its own colour in the app, taken from the Indian flag, so users always know which workspace they are in.

Role	Colour	Who they are	What they do
Operator	Saffron	Analysts and communication staff	Upload sources, choose outputs, edit drafts, send work for review, create emergency alerts
Reviewer	Green	Senior officers who hold a DSC token	Check outputs, add comments, approve and sign, send work back, withdraw old records
Admin	Navy blue	The system manager	Set up the app, manage users, templates, security rules, AI models, updates and the public verify page
Public (guest)	None	Any citizen, hospital, company or journalist	Only verify documents: scan a QR code, upload a file or paste a message. No login, no access to any data

4.1 What each role is allowed to do

Permission	Operator	Reviewer	Admin	Public
Verify a document (QR, file or message)	Yes	Yes	Yes	Yes
Create and edit content	Yes	No	No	No
Use emergency mode	Yes	Yes	No	No
Approve and sign	No	Yes	No	No
Withdraw (revoke) signed records	No	Yes	Yes	No
Manage users and templates	No	No	Yes	No
Change security rules	No	No	Yes	No
Update the public verify page	No	No	Yes	No

4.2 How the public reaches Pramaan AI without an account

The public never opens the main app. Instead, there is a **small separate verification page** (for example [verify.\[organisation\].gov.in](#)) hosted on a government server such as NIC or MeghRaj. It holds **only** the organisation's public certificate and the list of withdrawn documents. It contains no reports, no names and no AI.

When a citizen scans the QR code on a document, the phone opens this page. The QR code itself carries the document's fingerprint and digital signature, and the phone checks it against the public certificate. The

Admin keeps the page up to date by carrying a small signed update file from the offline office computer on a USB drive. Data only ever flows **out** of the office, never in.

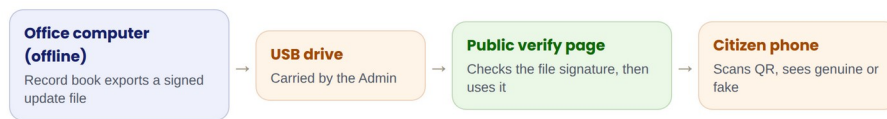


Figure 2: One-way flow from the offline office computer to the public verify page

5. Use cases: how each person uses the app

Each use case below is a short, real-life story showing who uses the app, what situation they are in, what they do step by step, and what they get. The sample names (Priya, Arjun, Kavya, Rahul, Meera) are used across the whole design.

5.1 Operator use cases

O1. Turning a threat report into a full advisory kit

Who: Priya, Operator, cyber operations team

Situation: A 30-page report on a ransomware attack against hospitals arrives at 9 am. Priya needs an advisory, a summary, slides and public posts in English, Hindi and Tamil.

What they do:

1. Opens **New transformation** and uploads the PDF.
2. Reads the **safety check**: it found a “RESTRICTED” marking and 6 internal IP addresses, and suggests TLP:AMBER. She accepts.
3. Picks the **Cyber advisory kit**, adds the X thread, selects English, हिन्दी and தமிழ், audience “IT and security teams” and tone “Formal”.
4. Clicks **Generate**. The summary and X thread appear in about 20 seconds; the rest follow within minutes.
5. Opens the advisory, clicks “42 hospitals...” and sees the matching line on page 4 of the report.
6. Clicks **Send for review**.

Result: About 30 minutes of work instead of 1–2 days. All versions state the same facts, and the public posts do not contain the internal IP addresses.

O2. A public awareness campaign

Who: Priya, Operator

Situation: A phishing scam using fake electricity bills is spreading. The goal is to warn ordinary citizens.

What they do:

1. Uploads the incident note and chooses the **Public awareness kit** (LinkedIn post, X thread, infographic, video package).
2. Sets audience “General public”, tone “Calm”, style “Simple”.
3. Previews the posts in the **Social posts** screen, where the character limits are already checked.
4. Picks the “Steps” layout for the **infographic** and downloads the PNG.
5. Listens to the **narrated video preview** in Hindi and adjusts one scene’s narration.

Result: A complete, consistent campaign in several languages, with a QR code on the infographic so people can check it is real.

O3. Emergency alert in all 22 languages

Who: Priya, Operator (on emergency duty)

Situation: Scam links pretending to be hospital bills are spreading fast. People need a warning right now, in their own languages.

What they do:

1. Clicks **Emergency alert** on the dashboard.
2. Chooses "Cyber fraud", severity "Warning", area "All states", and writes one short English message (under 160 characters).
3. The app instantly shows the alert in all 22 languages with voice announcements, and confirms the wording will not cause panic.
4. Clicks **Send for fast-track approval**. Any on-duty reviewer is notified at once.

Result: A signed, multilingual alert with voice versions within minutes, ready for SMS, radio or social media.

O4. Combining several different sources

Who: Priya, Operator

Situation: She has a PDF report, a photo of a printed circular and a Hindi voice recording of a briefing call.

What they do:

1. Drags all three into the upload screen.
2. The app reads the PDF, uses offline OCR on the photo, and turns the Hindi speech into text with IndicConformer.
3. All three are merged into **one fact sheet**.

Result: Outputs that combine everything, with each sentence still traceable to the exact source it came from.

O5. Starting from a prompt or voice, with no document

Who: Rahul, Operator

Situation: He has no report, only an idea: explain the new data protection rules to students.

What they do:

1. Chooses **Text or prompt** or presses **Speak** and describes what he needs in Hindi.
2. Picks outputs and settings as usual.

Result: Draft outputs created from his instructions. Because there is no source document, the app marks them clearly as needing a careful review.

O6. Fixing a sentence the source does not support

Who: Priya, Operator

Situation: The AI wrote "Attackers are believed to be demanding payment in cryptocurrency", but the report never says this.

What they do:

1. Sees the sentence highlighted in **yellow** with the note "1 sentence not found in source".
2. Chooses **Remove**, or **Keep as analyst note** if her own knowledge supports it.

Result: No unsupported claim goes out silently. The reviewer sees exactly what was changed.

O7. Updating when the incident grows (version 2)

Who: Priya, Operator

Situation: Two days later, a new report says 57 hospitals in six states are now affected.

What they do:

1. Adds the new report to the same job.
2. Opens **Version compare**: added text in green, removed text in red, plus a summary ("42 → 57 hospitals, 5 → 6 states, +1 action").
3. Regenerates all outputs from version 2 and sends it for review.

Result: A clear, trackable update. The old signed version is marked "Replaced by #0142" in the records.

O8. Watch folder: drafts ready before you arrive

Who: Priya, Operator

Situation: Reports arrive at all hours in a shared folder.

What they do:

1. Turns on **Watch folder**, points it to the incoming folder, picks a default kit and languages.
2. Each morning she opens the ready drafts, checks them and sends them for review.

Result: Routine reports are drafted automatically. Nothing is ever published without a human approving it.

O9. Work sent back by the reviewer

Who: Priya, Operator

Situation: Arjun sent back the flood advisory because a district name in the Tamil version was wrong.

What they do:

1. Sees "Sent back" on her dashboard with Arjun's comments on the exact lines.
2. Fixes the lines (or regenerates that part) and resends.

Result: Mistakes are fixed before release, and the whole conversation is kept in the audit trail.

5.2 Reviewer use cases

R1. Reviewing and signing a kit

Who: Arjun, Reviewer (holds a Class 3 DSC token)

Situation: Priya's ransomware kit is waiting in his review queue.

What they do:

1. Opens the kit and reads the **automatic checks**: sentences traced, facts matching, sensitive details hidden, safe public wording, no hidden tricks.
2. Reads each output and adds comments where needed.
3. Plugs in his **DSC token**, enters his PIN and confirms he has reviewed the work.
4. Clicks **Sign 21 files**.

Result: Every file is signed, gets a QR code and is recorded in the record book. Priya is notified that the Campaign Kit is ready.

R2. Fast-track approval of an emergency alert

Who: Arjun, Reviewer on duty

Situation: An emergency alert arrives, marked in red at the top of his queue.

What they do:

1. Opens it, checks the English message and a few translations, listens to one voice announcement.
2. Signs it immediately.

Result: Approval in minutes, without skipping the human check.

R3. Sending work back with notes

Who: Arjun, Reviewer

Situation: The Tamil flood advisory names a district that is not in the English source.

What they do:

1. Chooses reasons ("Facts need checking", "Language quality").
2. Adds comments on the exact lines and a short note to Priya.
3. Clicks **Send back**.

Result: Clear, specific feedback. Nothing wrong is signed.

R4. Withdrawing an old advisory

Who: Arjun, Reviewer

Situation: An older VPN advisory is now wrong and must not be trusted any more.

What they do:

1. Opens **Signed records**, finds the record and clicks **Revoke**.
2. The Admin includes it in the next public page update.

Result: Anyone who scans the old QR code now sees that it has been withdrawn.

5.3 Admin use cases

A1. Installing and setting up for the first time

Who: Kavya, Admin

Situation: A new office wants to start using Pramaan AI on a computer that has no internet.

What they do:

1. Runs the installer from the USB drive.
2. The **first-time setup** checks memory, graphics card and disk space, and confirms the Indian AI models are installed.
3. Creates the first Admin account and saves the 8 recovery codes offline.
4. Adds organisation details, the letterhead and the signing certificate.

Result: The app is ready in about 5 minutes after installation.

A2. Approving a new user

Who: Kavya, Admin

Situation: Rahul has filled in the “Request access” form.

What they do:

1. Sees “2 access requests” on the Admin overview.
2. Checks his details, confirms the role (Operator) and clicks **Approve**.

Result: Rahul can now sign in. The approval is recorded in the audit trail.

A3. Adding a reviewer who can sign

Who: Kavya, Admin

Situation: Meera is joining as a Reviewer and has a Class 3 DSC token.

What they do:

1. Opens **Add user**, fills in her details, picks **Reviewer**, turns on “Holds a Class 3 DSC token” and “Emergency duty”.
2. Shares the temporary password with her in person.

Result: Meera can review and sign from her first day.

A4. Investigating something suspicious

Who: Kavya, Admin

Situation: The system blocked a hidden instruction inside an uploaded PDF.

What they do:

1. Opens the **Audit trail**, filters by “Security” and sees who uploaded the file and when.
2. Exports the signed log for the investigation team.

Result: A complete, tamper-proof history of what happened.

A5. Running on a weaker computer

Who: Kavya, Admin

Situation: A branch office has computers with only 16 GB of memory.

What they do:

1. Opens **AI models** and switches to **Lite mode** (Param2 17B).

Result: The app works on smaller computers, with slightly simpler writing.

A6. Keeping the official look

Who: Kavya, Admin

Situation: Outputs must use the organisation’s letterhead and slide style.

What they do:

1. Uploads the letterhead in **Templates** and sets it as the default.

Result: Every document looks official automatically.

A7. Tightening security rules

Who: Kavya, Admin

Situation: A new classification word must always be detected.

What they do:

1. Opens **Security and policies**, adds the word, and checks the sharing rules table.

Result: The scanner catches the new word in every future upload.

A8. Updating the public verify page

Who: Kavya, Admin

Situation: Several documents were signed or withdrawn this week.

What they do:

1. Opens **Public verify page**, exports the signed update file to USB, and uploads it on the public server.

Result: The public always sees correct “genuine” and “withdrawn” results. Nothing from the internet touches the office computer.

A9. Updating the app and keeping backups

Who: Kavya, Admin

Situation: A new version arrives on a USB drive.

What they do:

1. Opens **Updates and backup**. The app checks the update’s signature, backs everything up and installs it.

Result: Safe updates with no internet, and encrypted backups every night.

A10. Checking the record book

Who: Kavya, Admin

Situation: She wants to be sure no signed record has been secretly changed.

What they do:

1. Opens **Record book** and clicks **Check the whole chain**.

Result: “Chain intact · 142 of 142 entries verified” in about 2 seconds.

5.4 Public use cases

P1. Checking a real advisory

Who: A hospital IT officer

Situation: An advisory arrived by email. Is it really from the agency?

What they do:

1. Scans the QR code with the phone camera.
2. The verify page opens (no app, no login).

Result: “Genuine”, with the issuing office, signer, date and “not changed since signing”.

P2. Catching a fake WhatsApp message

Who: A citizen

Situation: A forwarded “URGENT GOVT CYBER ALERT” asks for an OTP within 1 hour.

What they do:

1. Opens the verify page and pastes the message.

Result: “Not genuine”: no signature, suspicious link, asks for an OTP, pressure words. The page shows the national cybercrime helpline 1930.

P3. Spotting an edited copy

Who: A journalist

Situation: Someone changed one number in a real advisory and shared it.

What they do:

1. Uploads the file on the verify page.

Result: The fingerprint no longer matches the signature, so the result is “Not genuine: changed after signing”.

P4. Seeing that an advisory was withdrawn

Who: A company security team

Situation: They are still following an old advisory.

What they do:

1. Scan its QR code.

Result: “Withdrawn” with the date, so they know to look for the newer version.

5.5 Other organisations that can use Pramaan AI

Organisation	How they would use it
NTRO, NCIIPC, CERT-In, state CERTs	Threat advisories, briefings and public warnings
Police cyber cells	Scam warnings in local languages, awareness posts and videos
Disaster management (for example NDMA)	Flood, cyclone and heatwave alerts in all 22 languages with voice announcements
Health departments	Health advisories, outbreak updates and public awareness
Ministries and departments	Explaining new policies and rules to officers and citizens
Universities and training bodies	Turning research papers into summaries, slides and explainers

6. Complete list of features

Features are grouped by what they help with. Each one says what it does and why it matters.

A. Input: what the user can give

Feature	What it does	Why it matters
Text or prompt	Paste text or describe what you need	Works even without a document
Documents	PDF, Word, text and saved web pages	Covers most official material
Scanned pages and images	Reads text from images offline (OCR), including Indian scripts	Many government papers are scanned
Audio and video	Turns speech into text offline	Briefing calls and videos become usable
Voice instructions	Speak in your own language	Faster and more inclusive
Multiple sources	Up to 5 sources merged into one fact sheet	Real incidents rarely come from one file

B. The AI engine

Feature	What it does	Why it matters
Fact sheet	The AI reads once and writes down the key facts	All outputs agree with each other
7 output types	Advisory, executive summary, presentation, video package, infographic, LinkedIn post, X thread	Everything NTRO asked for
Many at once	Pick any combination; all are made from the same source	One click instead of seven jobs
Kits	Ready-made sets like "Cyber advisory kit"	Saves time on common jobs

C. Settings the user controls

Feature	What it does	Why it matters
Audience, tone, objective, style	For example: senior officers, calm, warn and instruct, official	The same facts, written for the right reader
Level of detail	Short, medium or detailed	Fits the time the reader has
Language	English plus 22 Indian languages	Reaches every citizen
Sharing level (TLP)	Red, Amber, Green or Clear	Controls who may see each

Feature	What it does	Why it matters
		output
Letterhead and template	Organisation's official look	Outputs look official

D. Trust and accuracy

Feature	What it does	Why it matters
Click-to-source	Click any sentence to see the matching line in the source	Anyone can check the AI's work
Unsupported-line warning	Sentences not found in the source turn yellow	Catches made-up facts
AI self-check	The AI re-reads its output against the fact sheet and fixes errors	Fewer mistakes before a human even looks
Fact match across outputs	Numbers, dates and names must agree everywhere	Consistent official communication
Quality score	Scores for tracing, readability, format rules and tone	Quick view of how good each output is
"AI-assisted, human-approved" label	Printed on every document	Honest and transparent

E. Security and privacy

Feature	What it does	Why it matters
Fully offline	No internet needed; nothing leaves the computer	Safe for secret material
Sensitive information scanner	Finds Aadhaar, PAN, phone numbers, emails, IP addresses, passwords and classification words	Warns before anything is shared
Audience-based auto-hiding	Internal outputs keep details; public outputs hide them automatically	Prevents accidental leaks
Hidden-instruction shield	Catches text inside files that tries to control the AI (prompt injection)	Stops attackers from tricking the AI
Encrypted storage	Database and files are locked with encryption	A stolen laptop reveals nothing
Roles and audit trail	Each person sees only what they need; every action is recorded	Accountability

F. Anti-fake protection

Feature	What it does	Why it matters
Digital signature (DSC)	Reviewers sign with an Indian Class 3 DSC USB token	Legally recognised official seal
QR code on every file	Links the document to its signature	Anyone can check it
Record book	Each signed document's fingerprint is linked to the one before it	Any secret change is exposed
Withdraw (revoke)	Mark old documents as withdrawn	People stop trusting outdated advice
"Is this real?" checker	Checks QR codes, files and pasted messages for signatures and scam signs	Catches fake alerts

G. Indian languages and voice

Feature	What it does	Why it matters
22 Indian languages	Translation with IndicTrans2	National reach
Indian voices	Narration and audio versions with Indic Parler-TTS	For videos, radio and visually impaired users
Voice input	Speak instead of typing	Easier for everyone

H. Smart extras

Feature	What it does	Why it matters
Emergency mode	Short alerts in all 22 languages with voice, fast-track review	Minutes instead of hours in a crisis
Public-release safety check	Flags panic-causing, offensive or unverified wording	Protects public trust
Cyber-intelligence extractor	Pulls CVE numbers, IP addresses and file fingerprints into a table	Advisories are ready for IT teams
Built-in editor	Quick buttons: Shorter, More formal, Simpler, Regenerate part	Easy fine-tuning
Version compare	Version 2 with changes highlighted	Clear updates as incidents grow
Watch folder	New reports in a folder are drafted automatically	Saves time on routine work
Campaign Kit	All files and languages in one signed package	Easy to share with other departments

I. Administration

Feature	What it does	Why it matters
Users and access requests	Approve, add, lock users; see what each role can do	Controlled access
AI models and performance	Full or Lite mode, model status, speed settings	Works on different computers
Templates	Letterheads, advisory, slide and infographic designs	Consistent official look
Security and policies	Scanner rules, sharing rules, encryption, certificates, sessions	One place to control safety
Public verify page settings	Export the signed update file for the public page	Keeps public checks correct
Offline updates and backup	Signed updates from USB; nightly encrypted backups	Safe maintenance without internet

J. Speed

Feature	What it does	Why it matters
Efficient Indian AI	Sarvam 30B uses only a small part of its “brain” for each word	Fast on normal computers
Read once, reuse	The fact sheet is reused for every output	Much less waiting
Quick mode and think mode	Deep thinking only for the fact sheet	Short posts are very fast
Live typing and short outputs first	Text appears as it is written	Users see progress immediately
Remember past results	The same file again gives instant results	No repeated work

7. App screens (design)

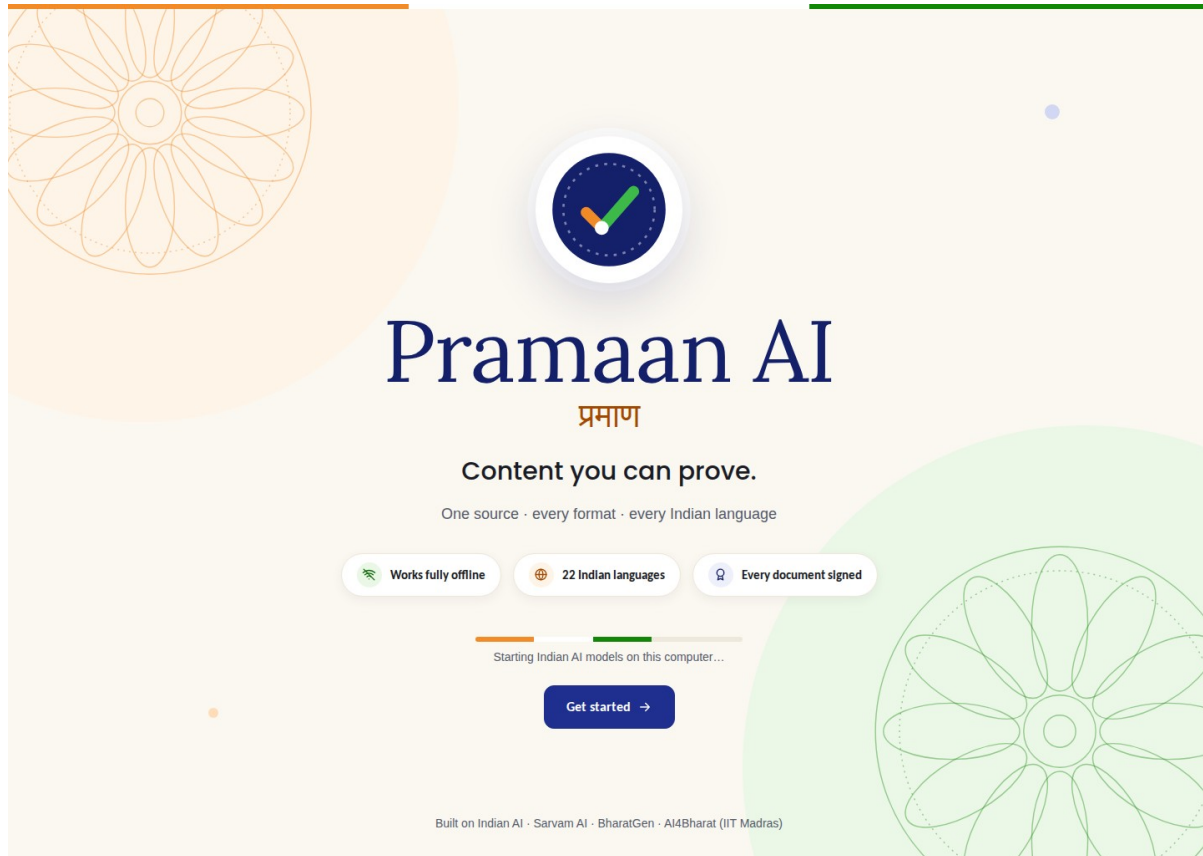
The full design has **46 screens**, covering every role from the splash screen to the last settings page. They are saved in the “**Designs**” folder next to this document, in three forms:

- **Screen images:** one picture per screen, grouped in folders by role.
- **All screens PDF:** every screen in one file, easy to share.
- **Clickable prototype:** open “index.html” in a web browser and click through the app. Buttons and menus move between screens.

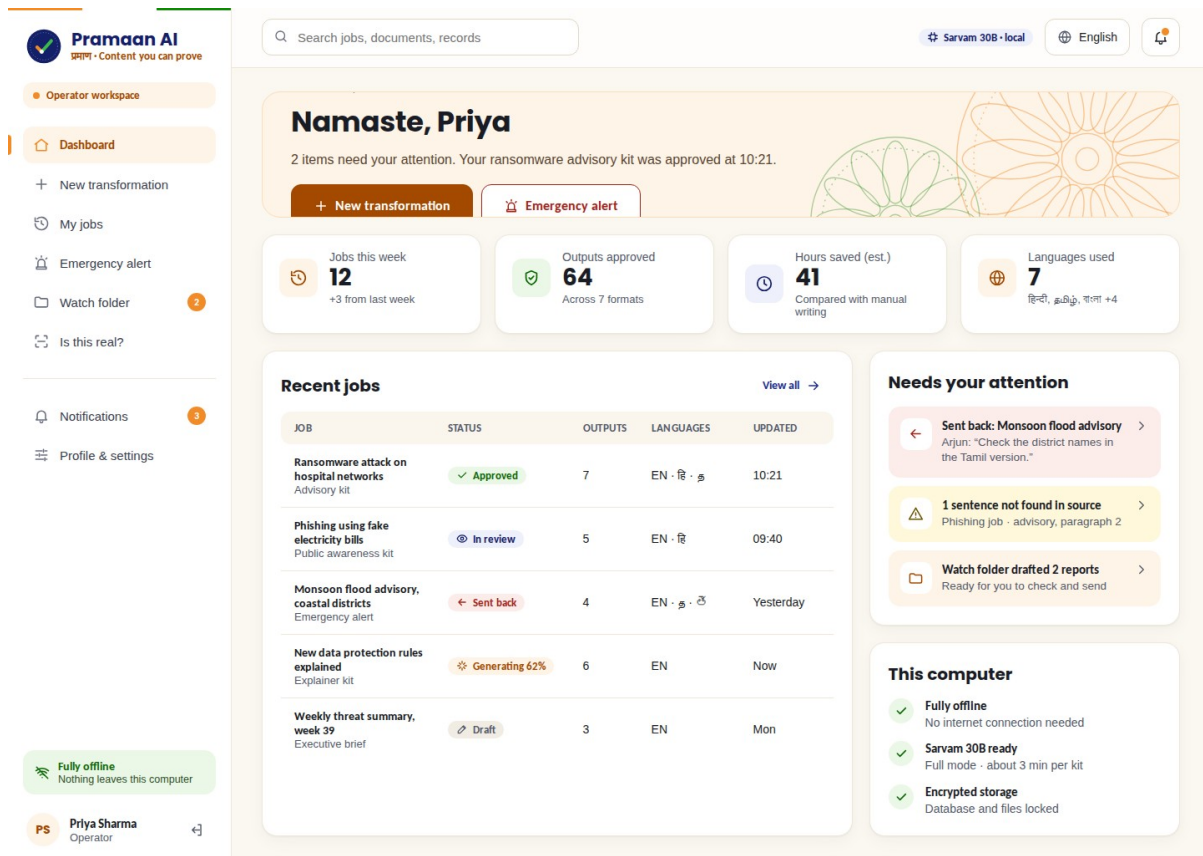
No.	Screen	Used by	What it shows
01	Splash screen	Everyone	Logo, name, tagline
02	Language selection	Everyone	All 22 Indian languages plus English
03	Sign in	Staff	Password or DSC token sign-in
04	Sign up (request access)	New staff	Request an Operator or Reviewer account
05	Forgot password	Staff	Reset through the Admin or a recovery code
06	Access request pending	New staff	Status of the request
07	First-time setup	Admin	Computer check, models, first Admin
08	Operator dashboard	Operator	Jobs, stats, items needing attention
09	New transformation 1: Add sources	Operator	Upload, kits, speak
10	New transformation 2: Safety check	Operator	Sensitive items, TLP, hidden tricks
11	New transformation 3: Outputs and settings	Operator	Choose outputs, languages and settings
12	Generating	Operator	Fact sheet, progress, live preview
13	Results: advisory with source trace	Operator	Click-to-source, yellow warnings, quality score
14	Results: presentation	Operator	Slides, notes, download
15	Results: video package	Operator	Scenes, narration, voices, subtitles
16	Results: social posts	Operator	LinkedIn post and X thread
17	Results: infographic	Operator	Layouts, key messages, PNG
18	Results: executive summary	Operator	One page, audio version
19	Campaign kit download	Operator	Signed package of all files
20	My jobs	Operator	History with filters

No.	Screen	Used by	What it shows
21	Version compare	Operator	Version 1 vs version 2
22	Emergency alert	Operator	22 languages with voice
23	Watch folder	Operator	Automatic drafts
24	Reviewer dashboard	Reviewer	Review queue and DSC status
25	Review a kit	Reviewer	Checks and line comments
26	Sign with DSC	Reviewer	Token, PIN, confirmation
27	Signed successfully	Reviewer	QR, fingerprint, next steps
28	Send back with notes	Reviewer	Reasons and comments
29	Signed records	Reviewer	Archive and revoke
30	Admin overview	Admin	Health, requests, security events
31	Users and access	Admin	Users, requests, permissions
32	Add or edit user	Admin	Role, DSC, temporary password
33	Audit trail	Admin	Every action, tamper-proof
34	AI models and performance	Admin	Full/Lite mode, models, speed
35	Templates	Admin	Letterheads and designs
36	Security and policies	Admin	Scanner, sharing rules, encryption
37	Record book	Admin	Chain of signed fingerprints
38	Public verification page	Admin	One-way update to the public page
39	Updates and backup	Admin	USB updates, backups
40	Notifications	All staff	Updates about jobs
41	Profile and settings	All staff	Language, accessibility, password
42	Is this real? (phone)	Public	Three ways to check
43	Scanning QR (phone)	Public	Camera view
44	Paste a message (phone)	Public	Check a forwarded message
45	Result: genuine (phone)	Public	Issuer, signer, date
46	Result: fake (phone)	Public	Reasons and helpline 1930

7.1 A few key screens



Screen 01: Splash screen



Screen 08: Operator dashboard

Pramaan AI
प्रमाण - Content you can prove

Operator workspace

- Dashboard
- New transformation
- My Jobs**
 - Emergency alert
 - Watch folder (2)
 - Is this real?
- Notifications (3)
- Profile & settings

Fully offline
Nothing leaves this computer

PS Priya Sharma
Operator

Search jobs, documents, records

Sarvam 30B · local English

JOB #0142 · 2 SOURCES · 3 LANGUAGES

Ransomware attack on hospital networks

6 of 7 ready TLP:AMBER Campaign kit Send for review

Advisory 94 Executive summary 96 Presentation 91 Video package 93 Social posts 96 Infographic 92

English हिन्दी தமிழ் Shorter More formal Simpler Regenerate part

ADVISORY · PRM-2026-0142 Severity: High TLP:AMBER

Ransomware campaign targeting hospital networks

Overview

A ransomware group is encrypting patient-record systems at hospitals after entering through an unpatched remote-access gateway. 42 hospitals across five states have reported disruption since 22 September. Attackers are believed to be demanding payment in cryptocurrency. Backups kept on the same network were also encrypted in several cases.

Indicators found in the source

Vulnerability	CVE-2026-XXXX (sample)
IP addresses	6 internal addresses - internal copy only
File fingerprint	SHA-256 9f2c_e41a (sample)

Recommended actions

1. Apply the vendor's September patch to every remote-access gateway today.
2. Block the listed addresses at the network edge and review connection logs.
3. Move backups offline and test that one restore works end to end.

Source trace Page 4
sample-ransomware-report.pdf

...the gateway flaw was first exploited on 22 September. As of 28 September, 42 hospitals in five states have reported disruption, mostly to patient-record and billing systems...

Open page 4

1 sentence not found in source
"Attackers are believed to be demanding payment in cryptocurrency."

Remove Keep as analyst note

Quality score 94

Traced to source	47 / 49
Readability	90
Format rules	100
Tone match	92

Screen 13: Results with source trace and the yellow "not found in source" warning

Pramaan AI
प्रमाण - Content you can prove

Reviewer workspace

- Review queue (4)
- Signed records
- Is this real?
- Notifications (3)
- Profile & settings

Fully offline
Nothing leaves this computer

AM Arjun Mehta
Reviewer

Search jobs, documents, records

Sarvam 30B · local English

REVIEW · JOB #0142 V2 · FROM PRIYA SHARMA

Ransomware attack on hospital networks

TLP:AMBER Compare with v1 Send back Approve & sign

7 outputs

- Advisory
- Executive summary
- Presentation
- Video package
- Infographic
- Linked
- X threat

EN

Sign with your DSC token
Your signature seals every file so no one can change it.

Class 3 DSC token detected
USB token - certificate read successfully ✓ Ready

Issued to Arjun Mehta

Issued by [LICENSED CERTIFYING AUTHORITY]

Certificate class Class 3 (signing)

Valid until [EXPIRY DATE]

You are signing 7 outputs × 3 languages = 21 files for job #0142 v2. Each file gets a QR code and an entry in the record book.

Token PIN

☒ I have reviewed these outputs and approve them for release

Cancel Sign 21 files

Automatic checks

- ✓ Every line traced to source
48 of 49 · 1 labelled analyst note
- ✓ Facts match across outputs
Numbers, dates, names agree in 3 languages
- ✓ Sensitive details hidden in public posts
6 IPs and 1 email removed
- ⚠ Public-release wording
1 phrase softened to avoid panic
- ✓ No hidden instructions in sources
- ✓ Sharing label applied
TLP:AMBER on every file

Screen 26: Reviewer signs with a DSC token


Pramaan AI
प्रमाण - Content you can prove

Admin workspace

- Overview
- Users & access 2
- Audit trail
- AI models
- Templates
- Security & policies
- Record book
- Public verify page**
- Updates & backup
- Notifications 3
- Profile & settings

Fully offline
Nothing leaves this computer

KN Kavya Nair
Admin

Sarvam 308 · local

English

ADMIN

Public verification page

A small public website where anyone can check a document. It holds no documents and no secrets.


verify.[organisation].gov.in
Hosted on a government server (for example NIC or MeghRaj) · last updated today, 10:35

Up to date


Genuine checks this week
318
Scanned by the public


Fake or edited caught
23
Shown as not genuine


Withdrawn records seen
5
Shown as withdrawn

Update the public page

→ One-way: office → public

- Export the update file**
Contains the public certificate and the list of withdrawn records. No documents, no personal data.


Export update file
pramaan-public-update-0930.sig
- Copy it to a USB drive**
This computer never connects to the internet, so the file travels by hand.
- Upload it on the public server**
The public page checks the file's signature before using it.


Nothing from the internet can ever reach this computer.

What the public page holds

Contains	Never contains
✓ Organisation public certificate	✗ Any document or report
✓ List of withdrawn records	✗ Names, emails or user accounts
✓ Fake-message checking rules	✗ AI models or private keys

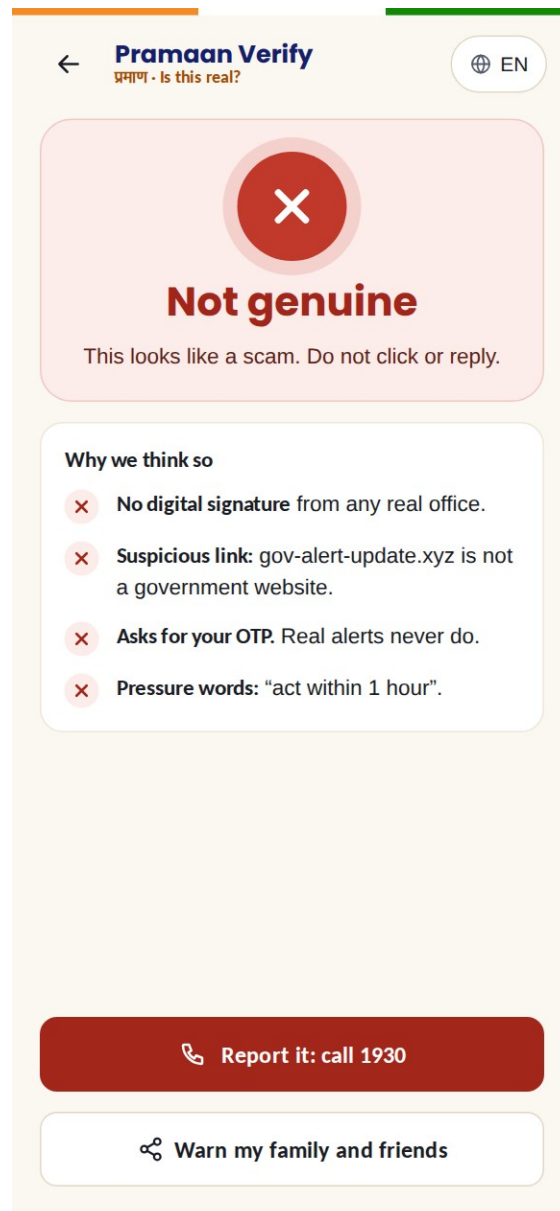
Fake-message rules

Run inside the visitor's browser. No message is uploaded.

- Look-alike or unofficial links ☒
- Requests for OTP, PIN or password ☒
- Pressure words like "act within 1 hour" ☒
- Show helpline 1930 on every fake result ☒

Screen 38: Admin keeps the public verify page up to date

Page 26



Screen 46: A citizen's phone showing a fake alert

8. Technology stack

8.1 The big picture

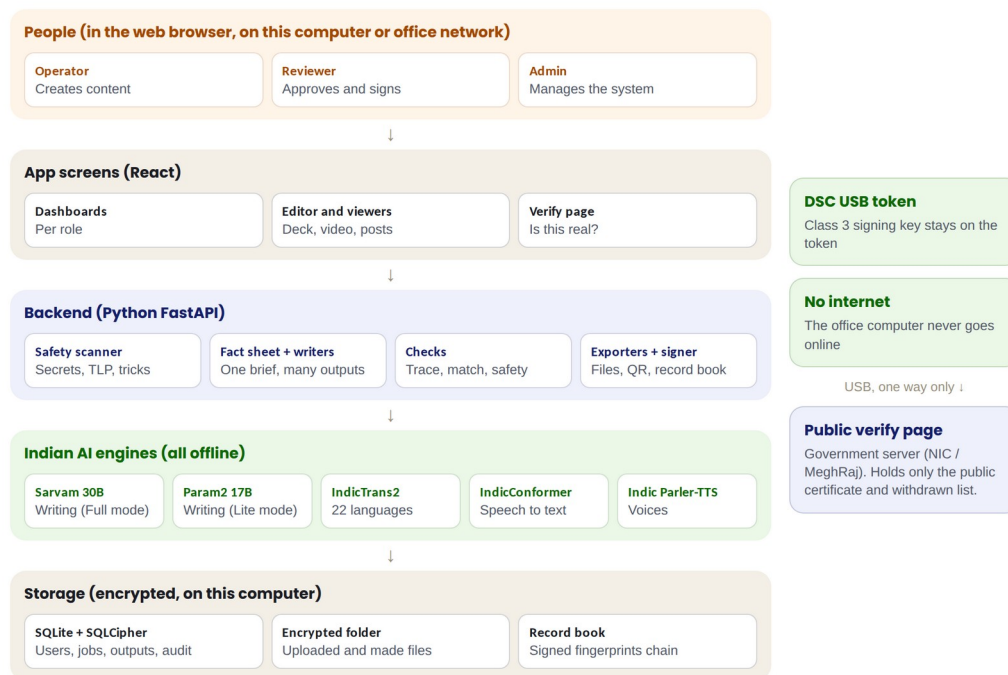


Figure 3: How the parts of Pramaan AI fit together

Think of it as five layers, from top to bottom: the **people** using a browser, the **screens**, the **backend** (the hidden “kitchen” where work happens), the **Indian AI engines**, and the **encrypted storage**. Everything sits on one computer (or one office server). Only a tiny public page lives outside, and it gets updates one way by USB.

8.2 Indian AI and language technology

Job	Technology	Made by	Notes
Main writing AI (Full mode)	Sarvam 30B	Sarvam AI, Bengaluru	Open source (Apache 2.0). Mixture-of-Experts: about 2.4B of 32B parameters active per word, so it is fast. 65K-token memory. Official offline version available.
Lite writing AI	BharatGen Param2 17B	BharatGen, IIT Bombay	For computers with less memory. Released under a BharatGen non-commercial licence, so check terms before any commercial use.
Translation (22 languages)	IndicTrans2	AI4Bharat, IIT Madras	Open source, runs offline, covers all 22 scheduled languages.
Speech to text (Indian languages)	IndicConformer	AI4Bharat, IIT Madras (Bhashini mission)	Runs offline, even on a normal processor.
Voice narration	Indic Parler-TTS	AI4Bharat, IIT	Natural Indian-language voices, offline.

Job	Technology	Made by	Notes
		Madras	
Digital signature	Class 3 DSC (CCA framework)	Indian certifying authorities	Signing key stays on a USB token.
Online mode (optional)	Bhashini, Sarvam APIs	MeitY, Sarvam AI	Only if an organisation allows internet; not needed for the core app.

8.3 App technology

Part	Technology	Why
Screens (frontend)	React with TypeScript	Modern, fast, reusable screens
Backend	Python with FastAPI	Python is the language of AI tools; FastAPI is simple and fast
Running the AI	llama.cpp server (or Ollama)	Runs Indian models offline, with 4-bit compression and prompt caching
Database	SQLite encrypted with SQLCipher (PostgreSQL for large offices)	A single encrypted file; no database server needed
File storage	Encrypted folder on disk	Uploads and outputs stay locked
Reading documents	PDF and Word readers; Tesseract OCR for scans	Handles every common input
English speech	Whisper (open source)	For English audio in videos
Making files	python-pptx, python-docx, PDF generator, Pillow	Real PowerPoint, Word, PDF and PNG files
Video preview	FFmpeg	Joins scenes, narration and subtitles into an MP4
Signing and fingerprints	SHA-256 hashing, PKCS#11 for the DSC token, QR code library	Standard, trusted security building blocks
Public verify page	Plain HTML and JavaScript (Web Crypto)	Tiny, fast, works on any phone, holds no secrets
Office network (optional)	Caddy web server with HTTPS	Secure access for a whole team

8.4 Why not Firebase any more

Our earlier projects used Firebase, which is Google's online cloud. A fully offline app cannot depend on any online service, so Firebase is replaced by a local encrypted database and local login. We still reuse our experience with role-based access, dashboards, voice features and the danger-word detector from our mental-health project, which becomes the sensitive information scanner.

8.5 An honest note on “everything Indian”

The AI brain, the language tools, the signature system and the standards are Indian. A few supporting tools are global open-source projects (llama.cpp, SQLite, Tesseract, Whisper, React, Python), because there are no mature Indian alternatives for them. These run offline on the user's computer, so data still never leaves India. This is the same approach Indian government systems use.

8.6 How we make it fast

- **Mixture-of-Experts model:** Sarvam 30B uses only a small part of its parameters for each word.
- **4-bit compression:** a smaller model file that runs faster with almost the same quality.
- **Read once, reuse (prompt caching):** the fact sheet is loaded into the AI's memory once and reused for every output.
- **Quick mode vs think mode:** deep reasoning only where it matters.
- **Keep the model loaded, stream text, short outputs first, remember past results.**
- **Lite mode** for weaker computers.

8.7 Main database tables

Table	What it stores
users	Name, employee ID, role, division, password (safely scrambled), DSC holder or not
jobs	Each transformation: title, owner, status, TLP level, version
sources	Details of uploaded files and their fingerprints (the files themselves are in the encrypted folder)
fact_sheets	Key facts with the exact source location of each fact
outputs	Each generated output, per language, with quality scores
reviews	Comments, approvals and send-backs
records	The record book: fingerprint, signer, time and the previous record's fingerprint
audit_log	Every action, linked in a chain so it cannot be edited
templates, settings	Letterheads, designs, security rules and preferences

8.8 Project folder structure

Folder	Contains
frontend/	React screens for all roles
backend/	FastAPI app: routes, safety scanner, fact sheet, writers, checks, exporters, signer
backend/prompts/	One prompt template per output type

Folder	Contains
models/	Downloaded Indian AI model files (not in Git)
verify-page/	The small public verification website
scripts/	install.sh, start.sh, update and backup scripts
docs/	README, architecture document, demo script

9. How Pramaan AI is downloaded, installed and used

9.1 Computer needed

	Full mode	Lite mode
Writing AI	Sarvam 30B	Param2 17B
Memory (RAM)	32 GB recommended	16 GB
Graphics card	Optional, makes it much faster	Optional
Free disk space	About 60 GB (estimate; the Sarvam 30B 4-bit file alone is roughly 20 GB)	About 35 GB (estimate)
Operating system	Linux (Ubuntu, BOSS Linux); macOS for development	Same

Important: These numbers are estimates. The team must test on its own laptops early, because model sizes and speeds vary.

9.2 Option A: install with internet (one time)

Open the terminal and type:

Command	What it does
<code>git clone https://github.com/[team]/pramaan-ai</code>	Downloads the app code
<code>cd pramaan-ai</code>	Opens the app folder
<code>./install.sh</code>	Installs everything, downloads the Indian AI models, creates the encrypted database and the first Admin

Internet is needed only for this one-time download.

9.3 Option B: fully offline install from USB

For secret offices with no internet at all, the app and the AI models are copied onto a USB drive or hard disk. On the office computer, run `./install.sh --offline`. Nothing is downloaded from the internet.

9.4 Everyday use

1. Type `./start.sh` in the terminal (or double-click the Pramaan AI icon).
2. The app opens in the web browser at `http://localhost:8080`. "localhost" means "this computer", so nothing goes online.
3. Sign in and work as normal.

9.5 For a whole office (office server)

Install Pramaan AI on one strong computer in the office. Team members open it from their own desks over the office network, for example at <https://pramaan.office.local>. A small web server (Caddy) adds a secure lock (HTTPS). The office still needs no internet.

9.6 Updates and backups

Updates arrive as a signed package on USB. The app checks the signature, backs everything up and then installs the update. Encrypted backups can run every night to a USB drive.

9.7 The public verify page

The small verify website is placed on a government server (for example NIC or MeghRaj). It contains only the public certificate and the withdrawn list. The Admin updates it with a signed file carried by USB. For the hackathon demo, the verify page can run from a laptop's Wi-Fi hotspot so judges can scan a QR code with their phones.

10. Security and privacy

Risk	How Pramaan AI protects against it
Secret reports leaking online	Fully offline; the office computer never connects to the internet
A stolen laptop	Database and files are encrypted; nothing is readable without the password
Secrets appearing in public posts	Sensitive information scanner plus automatic hiding in public outputs, controlled by TLP
AI making things up	Fact sheet, self-check, click-to-source and yellow warnings; a human always approves
Hidden instructions inside files (prompt injection)	Hidden-instruction shield scans text and images before the AI sees them
Fake government advisories	DSC signatures, QR codes, record book and the public “Is this real?” page
Someone changing a signed document	Its fingerprint stops matching; the verify page shows “Not genuine”
Someone secretly editing old records	Each record includes the previous record’s fingerprint, so the chain breaks
Stolen signing key	The key stays on the DSC USB token and needs a PIN
Misuse by an insider	Roles limit what each person can do; every action is in the audit trail

10.1 Laws and standards we follow

- **DPDP Act 2023** (India’s data protection law): personal data never leaves the device.
- **IndiaAI Mission**: uses India’s own AI models.
- **CCA framework**: Class 3 digital signature certificates.
- **CERT-In style** advisory structure.
- **TLP 2.0** (Traffic Light Protocol) sharing labels, used worldwide by security teams.
- **GIGW** (Guidelines for Indian Government Websites) for accessible, clear screens.

11. Build plan and demo

11.1 Build in this order

Phase	What we build	Done when
0. Setup	Accounts, laptops, download models, test Sarvam 30B speed	The model answers a test question on our laptop
1. Core engine	Upload → fact sheet → 7 outputs → file export	One report produces all 7 files
2. Trust	Click-to-source, self-check, fact match, quality score	Unsupported sentences turn yellow
3. Security	Scanner, TLP, auto-hiding, hidden-instruction shield	Public posts hide test IP addresses
4. Anti-fake	Signing, QR codes, record book, verify page, “Is this real?”	An edited copy shows “Not genuine”
5. Languages and voice	IndicTrans2, Indian voices, speech input	Hindi and Tamil outputs with narration
6. Roles and screens	Login, three roles, dashboards, admin screens	The full flow works end to end
7. Extras	Emergency mode, watch folder, version compare, templates	Only if time allows
8. Deliverables	README, 2-page architecture document, 2-minute video, 5 slides	Everything submitted

Rule: Build the core first and make it work perfectly. Anything unfinished goes on a “Future roadmap” slide. Judges see that as vision, not missing work.

11.2 The 2-minute demo

1. **Hook (0–10 s):** turn Wi-Fi off on screen, drop in a real public CERT-In advisory, tick all outputs.
2. **Speed:** summary and posts appear in seconds; show the deck, the video preview and the Hindi version.
3. **Truth:** click a sentence to show its source; show a yellow warning.
4. **Safety:** paste text with “RESTRICTED” and an IP address; show the warning and the hidden details in the public post.
5. **Proof:** sign with the DSC token, scan the QR with a phone: “Genuine”. Change one word: “Not genuine”.
6. **Close:** paste a fake WhatsApp alert into “Is this real?”: caught.

Always record a perfect “golden run” before the demo, and keep it ready in case anything goes wrong live.

12. Honest limitations and future roadmap

12.1 Limitations to be aware of

- Offline AI on a laptop is slower than cloud AI; a full kit may take a few minutes.
- Local models write slightly less fluently than the biggest global models; the fact sheet, self-check and human review make up for this.
- Voice quality and coverage must be tested for each Indian language.
- The public verify page is only as up to date as the last USB update.
- Translations of sensitive messages should always be checked by a native speaker before release.

12.2 Future roadmap

- Anchor record-book fingerprints to a permissioned national blockchain.
- A mobile app for QR verification.
- Training a custom model on an organisation's past advisories.
- Direct publishing to official portals after approval.
- Detection of deepfake images and videos.

13. Simple glossary

Term	Simple meaning
AI model / LLM	The “brain” that reads and writes text
Offline / air-gapped	Works with no internet connection at all
Fact sheet	Our list of key facts that every output is written from
Hallucination	When AI confidently writes something false
Grounding / source trace	Proving where each sentence came from
Prompt injection	Hidden text in a file that tries to trick the AI
PII	Personal information such as Aadhaar, PAN, phone numbers
TLP	Traffic Light Protocol: Red, Amber, Green, Clear labels for who may see something
Hash / fingerprint	A unique code for a file; changes completely if one character changes
Record book / hash chain	A list where each entry contains the previous entry's fingerprint, so secret edits are exposed
DSC	Digital Signature Certificate, an official digital seal kept on a USB token
QR code	A square code a phone camera can read
OCR	Reading text from a picture of a page
TTS / STT	Text-to-speech (computer speaks) / speech-to-text (computer types what you say)
Mixture-of-Experts (MoE)	An AI design that uses only a small part of its brain for each word, so it runs faster
4-bit compression (quantization)	Making the AI file smaller so it runs faster on normal computers
localhost	“This computer”: the app runs on your own machine
API key	A password that lets a program use an online service